

## JEE Mains 2019 Chapter wise Question Bank

## Parabola – Questions

Q1

Axis of a parabola lies along x-axis. If its vertex and focus are at distance 2 and 4 respectively from the origin, on the positive x-axis then which of the following points does not lie on it?

- (1)  $(5, 2\sqrt{6})$  (2)  $(8, 6)$   
(3)  $(6, 4\sqrt{2})$  (4)  $(4, -4)$

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Q2

If  $\theta$  denotes the acute angle between the curves,  $y = 10 - x^2$  and  $y = 2 + x^2$  at a point of their intersection, then  $|\tan \theta|$  is equal to:

- (1)  $\frac{4}{9}$  (2)  $\frac{8}{15}$   
(3)  $\frac{7}{17}$  (4)  $\frac{8}{17}$

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Q3

Two cards are drawn successively with replacement from a well-shuffled deck of 52 cards. Let X denote the random variable of number of aces obtained in the two drawn cards. Then  $P(X = 1) + P(X = 2)$  equals:

- (1)  $49/169$  (2)  $52/169$   
(3)  $24/169$  (4)  $25/169$

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Q4

Let A  $(4, -4)$  and B  $(9, 6)$  be points on the parabola,  $y^2 = 4x$ . Let C be chosen on the arc AOB of the parabola, where O is the origin, such that the area of  $\triangle ACB$  is maximum. Then, the area (in sq. units) of  $\triangle ACB$ , is:

- (1)  $31\frac{1}{4}$  (2)  $30\frac{1}{2}$   
(3) 32 (4)  $31\frac{3}{4}$

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Q5

If the parabolas  $y^2 = 4b(x - c)$  and  $y^2 = 8ax$  have a common normal, then which one of the following is a valid choice for the ordered triad  $(a, b, c)$ ?

- (1)  $\left(\frac{1}{2}, 2, 3\right)$  (2)  $(1, 1, 3)$   
(3)  $\left(\frac{1}{2}, 2, 0\right)$  (4)  $(1, 1, 0)$

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Q6

The length of the chord of the parabola  $x^2 = 4y$  having equation  $x - \sqrt{2}y + 4\sqrt{2} = 0$  is:

- (1)  $3\sqrt{2}$  (2)  $2\sqrt{11}$   
(3)  $8\sqrt{2}$  (4)  $6\sqrt{3}$

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Q7

## Parabola

If the area of the triangle whose one vertex is at the vertex of the parabola,  $y^2 + 4(x - a^2) = 0$  and the other two vertices are the points of intersection of the parabola and  $y$ -axis, is 250 sq. units, then a value of 'a' is :

- (1)  $5\sqrt{5}$  (2)  $5(2^{1/3})$   
(3)  $(10)^{2/3}$  (4) 5

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### Q8

The equation of a tangent to the parabola,  $x^2 = 8y$ , which makes an angle  $\theta$  with the positive direction of  $x$ -axis, is :

- (1)  $y = x \tan \theta + 2 \cot \theta$   
(2)  $y = x \tan \theta - 2 \cot \theta$   
(3)  $x = y \cot \theta + 2 \tan \theta$   
(4)  $x = y \cot \theta - 2 \tan \theta$

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### Q9

The tangent to the parabola  $y^2 = 4x$  at the point where it intersects the circle  $x^2 + y^2 = 5$  in the first quadrant, passes through the point :

- (1)  $\left(-\frac{1}{3}, \frac{4}{3}\right)$  (2)  $\left(\frac{1}{4}, \frac{3}{4}\right)$   
(3)  $\left(\frac{3}{4}, \frac{7}{4}\right)$  (4)  $\left(-\frac{1}{4}, \frac{1}{2}\right)$

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### Q10

If one end of a focal chord of the parabola,  $y^2 = 16x$  is at (1, 4), then the length of this focal chord is:

- (1) 25 (2) 22 (3) 24 (4) 20

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### Q11

The area (in sq. units) of the smaller of the two circles that touch the parabola,  $y^2 = 4x$  at the point (1, 2) and the  $x$ -axis is:

- (1)  $8\pi(2 - \sqrt{2})$  (2)  $4\pi(2 - \sqrt{2})$   
(3)  $4\pi(3 + \sqrt{2})$  (4)  $8\pi(3 - 2\sqrt{2})$

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### Q12

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If the tangent to the parabola  $y^2 = x$  at a point  $(\alpha, \beta)$ , ( $\beta > 0$ ) is also a tangent to the ellipse,  $x^2 + 2y^2 = 1$ , then  $\alpha$  is equal to:

- (1)  $\sqrt{2} - 1$  (2)  $2\sqrt{2} - 1$   
(3)  $2\sqrt{2} + 1$  (4)  $\sqrt{2} + 1$

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### Q13

If the line  $ax + y = c$ , touches both the curves  $x^2 + y^2 = 1$  and  $y^2 = 4\sqrt{2}x$ , then  $|c|$  is equal to

- (1) 2 (2)  $\frac{1}{\sqrt{2}}$   
(3)  $\frac{1}{2}$  (4)  $\sqrt{2}$

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### Q14

Let P be the point of intersection of the common tangents to the parabola  $y^2 = 12x$  and hyperbola  $8x^2 - y^2 = 8$ . If S and S' denote the foci of the hyperbola where S lies on the positive  $x$ -axis then P divides SS' in a ratio :

- (1) 13 : 11 (2) 14 : 13 (3) 5 : 4 (4) 2 : 1

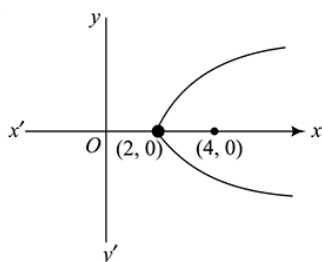
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## Parabola - Solutions

Q1

(2) Since, vertex and focus of given parabola is (2, 0) and (4, 0) respectively



Then, equation of parabola is

$$(y - 0)^2 = 4 \times 2(x - 2) \\ \Rightarrow y^2 = 8x - 16$$

Hence, the point (8, 6) does not lie on given parabola.

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Q2

(2) Since, the equation of curves are

$$y = 10 - x^2 \quad \dots(1)$$

$$y = 2 + x^2 \quad \dots(2)$$

Adding eqn (1) and (2), we get

$$2y = 12 \Rightarrow y = 6$$

Then, from eqn (1)

$$x = \pm 2$$

Differentiate equation (1) with respect to x

$$\frac{dy}{dx} = -2x \Rightarrow \left(\frac{dy}{dx}\right)_{(2,6)} = -4 \text{ and } \left(\frac{dy}{dx}\right)_{(-2,6)} = 4$$

Differentiate equation (2) with respect to x

$$\frac{dy}{dx} = 2x \Rightarrow \left(\frac{dy}{dx}\right)_{(2,6)} = 4 \text{ and } \left(\frac{dy}{dx}\right)_{(-2,6)} = -4$$

$$\text{At } (2, 6) \tan \theta = \left(\frac{(-4) - (4)}{1 + (-4) \times (4)}\right) = \frac{-8}{15}$$

$$\text{At } (-2, 6), \tan \theta = \frac{(4) - (-4)}{1 + (4)(-4)} = \frac{8}{-15}$$

$$\therefore |\tan \theta| = \frac{8}{15}$$

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Q3

(4)  $X$  = number of aces drawn

$$\therefore P(X=1) + P(X=2)$$

$$= \left\{ \frac{4}{52} \times \frac{48}{52} + \frac{48}{52} \times \frac{4}{52} \right\} + \left\{ \frac{4}{52} \times \frac{4}{52} \right\}$$

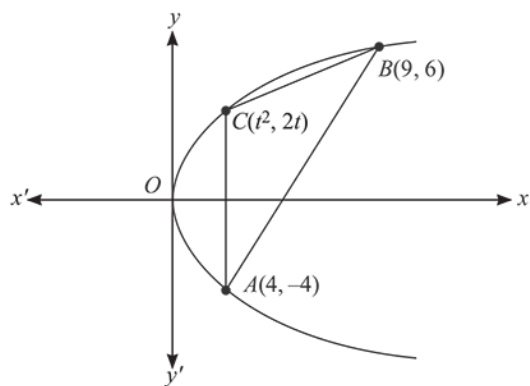
$$= \frac{24}{169} + \frac{1}{169} = \frac{25}{169}$$

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Q4

## Parabola

(1)



Let the coordinates of  $C$  is  $(t^2, 2t)$ .

Since, area of  $\triangle ACB$

$$= \frac{1}{2} \begin{vmatrix} t^2 & 2t & 1 \\ 9 & 6 & 1 \\ 4 & -4 & 1 \end{vmatrix}$$

$$= \frac{1}{2} |t^2(6+4) - 2t(9-4) + 1(-36-24)|$$

$$= \frac{1}{2} |10t^2 - 10t - 60|$$

$$= 5|t^2 - t - 6|$$

$$= 5 \left| \left( t - \frac{1}{2} \right)^2 - \frac{25}{4} \right| \quad [\text{Here, } t \in (0, 3)]$$

For maximum area,  $t = \frac{1}{2}$

Hence, maximum area  $= \frac{125}{4} = 31\frac{1}{4}$  sq. units

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(1, 2, 3, 4)

Normal to  $y^2 = 8ax$  is

$$y = mx - 4am - 2am^3 \quad \dots(1)$$

and normal to  $y^2 = 4b(x - c)$  with slope  $m$  is

$$y = m(x - c) - 2bm - bm^3 \quad \dots(2)$$

Since, both parabolas have a common normal.

$$\therefore 4am + 2am^3 = cm + 2bm + bm^3$$

$$\Rightarrow 4a + 2am^2 = c + 2b + bm^2 \text{ or } m = 0$$

$$\Rightarrow 4a - c - 2b = (b - 2a)m^2$$

or ( $X$ -axis is common normal always)

Since,  $x$ -axis is a common normal. Hence all the options are correct for  $m = 0$ .

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Q6

Q5

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## Parabola

(4) Let intersection points be  $P(x_1, y_1)$  and  $Q(x_2, y_2)$

The given equations

$$x^2 = 4y \quad \dots(1)$$

$$x - \sqrt{2}y + 4\sqrt{2} = 0 \quad \dots(2)$$

Use eqn (1) in eqn (2)

$$x - \sqrt{2} \frac{x^2}{4} + 4\sqrt{2} = 0$$

$$\sqrt{2}x^2 - 4x - 16\sqrt{2} = 0$$

$$x_1 + x_2 = 2\sqrt{2}, x_1x_2 = -16, (x_1 - x_2)^2 = 8 + 64 = 72$$

Since, points P and Q both satisfy the equations (2), then

$$x_1 - \sqrt{2}y_1 + 4\sqrt{2} = 0$$

$$x_1 - \sqrt{2}y_2 + 4\sqrt{2} = 0$$

$$(x_2 - x_1) = \sqrt{2}(y_2 - y_1) \Rightarrow (x_2 - x_1)^2 = 2(y_2 - y_1)^2$$

$$\Rightarrow PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(x_2 - x_1)^2 + \frac{(x_2 - x_1)^2}{2}}$$

$$= |x_2 - x_1| \cdot \frac{\sqrt{3}}{\sqrt{2}} = 6\sqrt{2} \times \frac{\sqrt{3}}{\sqrt{2}} = 6\sqrt{3}$$

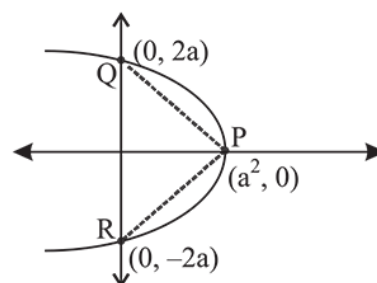
Hence, length of chord =  $6\sqrt{3}$ .

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Q7

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$$(4) y^2 = -4(x - a^2)$$



$$\begin{aligned} \text{Area} &= \frac{1}{2} (4a)(a^2) \\ &= 2a^3 \end{aligned}$$

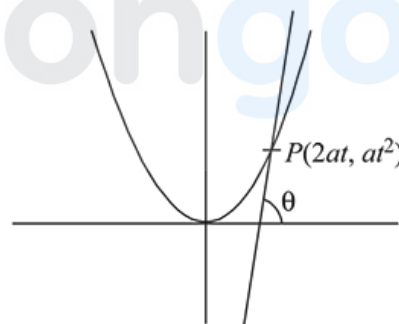
$$\text{Since } 2a^3 = 250$$

$$\Rightarrow a = 5$$

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Q8

$$(3) x^2 = 8y$$



Then, equation of tangent at P

$$tx = y + at^2$$

$$\Rightarrow y = tx - at^2$$

Then, slope  $t = \tan \theta$

$$\text{Now, } y = \tan \theta x - 2 \tan^2 \theta$$

$$\Rightarrow \cot \theta y = x - 2 \tan \theta$$

$$x = y \cot \theta + 2 \tan \theta$$

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## Parabola

**Q9**

- (3) To find intersection point of  $x^2 + y^2 = 5$  and  $y^2 = 4x$ , substitute  $y^2 = 4x$  in  $x^2 + y^2 = 5$ , we get  
 $x^2 + 4x - 5 = 0 \Rightarrow x^2 + 5x - x - 5 = 0$   
 $\Rightarrow x(x+5) - 1(x+5) = 0$   
 $\therefore x = 1, -5$   
 Intersection point in 1<sup>st</sup> quadrant be (1, 2).  
 Now, equation of tangent to  $y^2 = 4x$  at (1, 2) is  
 $y \times 2 = 2(x+1) \Rightarrow y = x+1$   
 $\Rightarrow x - y + 1 = 0 \quad \dots(i)$   
 Hence,  $\left(\frac{3}{4}, \frac{7}{4}\right)$  lies on (i)

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**Q10**

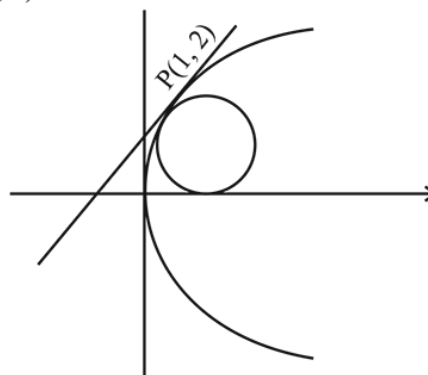
- (1)  $\because y^2 = 16x$   
 $\Rightarrow a = 4$   
 One end of focal of the parabola is at (1, 4)  
 $\because y$  - coordinate of focal chord is  $2at$   
 $\therefore 2at = 4$   
 $\Rightarrow t = \frac{1}{2}$   
 Hence, the required length of focal chord  
 $= a\left(t + \frac{1}{t}\right)^2 = 4 \times \left(2 + \frac{1}{2}\right)^2 = 25$

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**Q11**

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- (4) The circle and parabola will have common tangent at P (1, 2).



So, equation of tangent to parabola is,

$$y \times (2) = \frac{4(x+1)}{2} \Rightarrow 2y = 2x + 2 \Rightarrow y = x + 1$$

Let equation of circle (by family of circles) is  
 $(x - x_1)^2 + (y - y_1)^2 + \lambda T = 0$

$$\Rightarrow c \equiv (x-1)^2 + (y-2)^2 + \lambda(x-y+1) = 0$$

$\therefore$  circles touches  $x$ -axis.

$\therefore y$ -coordinate of centre = radius

$$\Rightarrow c = x^2 + y^2 + (\lambda - 2)x + (-\lambda - 4)y + (\lambda + 5) = 0$$

$$\frac{\lambda + 4}{2} = \sqrt{\left(\frac{\lambda - 2}{2}\right)^2 + \left(\frac{-\lambda - 4}{2}\right)^2} - (\lambda + 5)$$

$$\Rightarrow \frac{\lambda^2 - 4\lambda + 4}{4} = \lambda + 5 \Rightarrow \lambda^2 - 4\lambda + 4 = 4\lambda + 20$$

$$\Rightarrow \lambda^2 - 8\lambda - 16 = 0 \Rightarrow \lambda = 4 \pm 4\sqrt{2}$$

$$\Rightarrow \lambda = 4 - 4\sqrt{2} \quad (\because \lambda = 4 + 4\sqrt{2} \text{ forms bigger circle})$$

Hence, centre of circle  $(2\sqrt{2} - 2, 4 - 2\sqrt{2})$  and radius  
 $= 4 - 2\sqrt{2}$

$$\therefore \text{area} = \pi(4 - 2\sqrt{2})^2 = 8\pi(3 - 2\sqrt{2})$$

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**Q12**

## Parabola

- (4) Let tangent to parabola at point  $\left(\frac{1}{4m^2}, -\frac{1}{2m}\right)$  is

$$y = mx + \frac{1}{4m}$$

and tangent to ellipse is,  $y = mx \pm \sqrt{m^2 + \frac{1}{2}}$

Now, condition for common tangency,

$$\frac{1}{4m} = \pm \sqrt{m^2 + \frac{1}{2}} \Rightarrow \frac{1}{16m^2} = m^2 + \frac{1}{2}$$

$$\Rightarrow 16m^4 + 8m^2 - 1 = 0 \Rightarrow m^2 = \frac{-8 \pm \sqrt{64 + 64}}{2(16)}$$

$$= \frac{-8 \pm 8\sqrt{2}}{2(16)} = \frac{\sqrt{2} - 1}{4}$$

$$\alpha = \frac{1}{4m^2} = \frac{1}{4 \frac{\sqrt{2} - 1}{4}} = \sqrt{2} + 1$$

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### Q13

- (4) Equation of tangent on  $y^2 = 4\sqrt{2}x$  is  $yt = x + \sqrt{2}t^2$

This is also tangent on circle

$$\therefore \left| \frac{\sqrt{2}t^2}{\sqrt{1+t^2}} \right| = 1 \Rightarrow 2t^4 = 1 + t^2 \Rightarrow t^2 = 1$$

Hence, equation is  $\pm y = x + \sqrt{2} \Rightarrow |c| = \sqrt{2}$

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### Q14

## JEE Mains 2019 Chapter wise Question Bank

- (3) Equation of tangent to  $y^2 = 12x$  is  $y = mx + \frac{3}{m}$   
Equation of tangent to

$$\frac{x^2}{1} - \frac{y^2}{8} = 1 \text{ is } y = mx \pm \sqrt{m^2 - 8}$$

$\therefore$  parabola and hyperbola have common tangent.

$$\therefore \frac{3}{m} = \pm \sqrt{m^2 - 8} \Rightarrow \frac{9}{m^2} = m^2 - 8$$

Put  $m^2 = u$

$$u^2 - 8u - 9 = 0 \Rightarrow u^2 - 9u + u - 9 = 0$$

$$\Rightarrow (u + 1)(u - 9) = 0$$

$$\therefore u = m^2 \geq 0 \Rightarrow u = m^2 = 9 \Rightarrow m = \pm 3$$

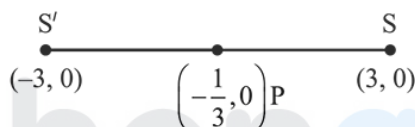
$\therefore$  equation of tangent is  $y = 3x + 1$

or  $y = -3x - 1$

$\therefore$  intersection point is  $P\left(-\frac{1}{3}, 0\right)$ .

$$e = \sqrt{1 + \frac{b^2}{a^2}} \Rightarrow e = \sqrt{1 + \frac{8}{1}} \Rightarrow e = 3$$

$\therefore$  foci  $(\pm 3, 0)$



$$\frac{SP}{SP'} = \frac{3 + \frac{1}{3}}{3 - \frac{1}{3}} = \frac{10}{8} = \frac{5}{4}$$

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